Practice Quiz: The Creative System

Part A: Multiple Choice

1. In Active Inference terms, the boundary between a creative system and its environment is called: A) The prediction error surface B) The Markov blanket C) The free energy gradient D) The affordance threshold

2. Why did Edison's Menlo Park laboratory produce so many inventions? A) Edison was the sole genius and worked alone B) The lab had unlimited funding and no constraints C) The lab was designed with a rich sensory surface and diverse information inputs D) The lab focused exclusively on a single technology domain

3. According to the Active Inference framework, constraints in a creative system function as: A) Obstacles that must be removed for creativity to flourish B) Priors in the generative model that focus creative search C) Random noise that interferes with the creative signal D) External forces that have no effect on internal creative processes

4. A "rugged" free energy landscape in a creative environment is most conducive to: A) Incremental improvements to existing products B) Meeting tight project deadlines C) Radical innovation and unexpected discoveries D) Systematic optimization of known parameters

5. The "adjacent possible" in invention refers to: A) The set of inventions that are one combinatorial step from current knowledge B) The nearest patent in a prior art search C) The competitor's most recent product D) The next scheduled milestone in a project plan

6. In the Shenzhen hardware ecosystem, what system-level feature most accelerates invention? A) Central government planning of all manufacturing activities B) Geographic concentration of suppliers reducing perception-action loop cycle time C) Strict intellectual property enforcement preventing copying D) Standardization on a single product category

7. Self-organization in creative systems (like the emergence of innovation clusters) occurs because: A) Government agencies deliberately create them through policy B) A single visionary leader coordinates all participants C) Reinforcing feedback loops drive the system toward low-free-energy attractors D) Market forces always produce geographically concentrated industry

Part B: Short Answer / Analysis

1. A startup has three engineers working from home in different cities. Their creative system has wide sensory inputs (internet, diverse backgrounds) but very slow feedback loops (they share work via email once per week). Using Active Inference concepts, diagnose the problem and propose two specific modifications to their creative system that would improve inventive output.

2. Consider a hackathon with a 48-hour time constraint and a theme ("build something that helps elderly people stay connected"). Analyze how these constraints function as priors. What regions of the solution space do they make more likely? What regions do they make less likely? Is this focusing effect beneficial for the participants' creative output?

3. You are designing a maker space for a community center. The space will serve inventors from diverse backgrounds — artists, engineers, hobbyists, students. Using the Active Inference concepts from this module, specify three design decisions you would make regarding (a) the physical boundary of the space, (b) the tools and materials provided, and (c) the social structures for interaction. Justify each decision in terms of Markov blankets, affordances, or feedback loops.